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Digital Discoveries introduces students to computer science as a vehicle for problem-solving, communication, and personal expression. The course focuses on the visible aspects of computing and computer science and encourages students to see where computer science exists around them and how they can engage with it as a tool for exploration and expression.

In Grade 8 Digital Discoveries, instructional time will emphasize the following areas:

Active learning experiences, relevant to students' lives, and provide students with authentic choices. Students are encouraged to be curious, solve personally relevant problems, and express themselves through creation. Learning is inherently social, so the course is designed to interweave lessons with discussions, presentations, peer feedback, and shared reflections. As students proceed through the pathway, the structures increasingly shift responsibility to students to formulate their own questions, develop their own solutions, and critique their work.

Diversifying the technological workforce is also critical. Addressing inequities in computer science is critical to making it accessible to all students. The tools and strategies in this course will help teachers understand and address well-known equity gaps within the field. All students can succeed in computer science when given the proper support and opportunities, regardless of prior knowledge.

M/J Grade 8 Digital Discoveries (#0200023)

Course Standards

Visit the specific benchmark webpage to find related instructional resources.

- [SC.8.CC.1.1](#): Design a digital product.

Clarifications:

Clarification 1: Instruction includes creating a product pertaining to the real world using a variety of digital tools and resources.

Clarification 2: Instruction includes the creation of a digital product that provides value to society individually or collaboratively.

- [SC.8.CO.1.1](#): Integrate information from multiple file formats into a single artifact.
- [SC.8.CO.1.2](#): Create a collaborative project utilizing an online digital application.

Clarifications:

Clarification 1: Projects include those that inform, persuade and entertain others.

- [SC.8.CS.2.1](#): Evaluate security and privacy issues that relate to computer networks and Internet of Things (IoT) devices.

Clarifications:

Clarification 1: Devices include phones, tablets, smartwatches and other emerging technologies.

- [SC.8.CS.2.2](#): Describe security and privacy issues that relate to computer networks.
- [SC.8.CS.2.3](#): Describe the permanency of data on the Internet, online identity and personal privacy.

Clarifications:

Clarification 1: Instruction includes discussing the “Right to be Forgotten.”

- [SC.8.ET.1.1](#): Identify the emerging features of mobile devices, smart devices and vehicles.
- [SC.8.ET.1.2](#): Identify challenges faced by users when learning to use computer interfaces.

- [SC.8.ET.1.3:](#) Identify the impact of natural resources on the manufacturing of computer hardware components.
- [SC.8.ET.1.4:](#) Analyze the increasing impact of access to the Internet on daily life.
- [SC.8.ET.2.2:](#) Discuss the utilization of intelligent behavior in technology.

Clarifications:

Clarification 1: Instruction includes speech and language understanding and computer vision.

Examples:

Example: Discuss the autonomous robotic vacuum cleaner's ability to map and analyze structures to avoid obstacles.

Example: Analyze the advantages of implementing drones to spray crops or detect predators in an agricultural environment.

- [SC.8.ET.3.1:](#) Investigate the advancement of robotics.
- [SC.8.HS.1.1:](#) Describe the impacts of the presence of technology and the lack of technology on everyday life.

Clarifications:

Clarification 1: Instruction includes comparing how the presence of technology has impacted our daily lives and how the lack of technology has impacted our daily lives.

- [SC.8.HS.2.3:](#) Investigate the causes of physical body changes due to device usage.

Clarifications:

Clarification 1: Focal points include research on muscle, nervous and bone systems.

Clarification 2: Instruction is limited to the physical effects of direct digital device usage.

- [SC.8.HS.3.2:](#) Analyze how digital media and communication influence behavior.

Clarifications:

Clarification 1: Instruction includes the influences on individuals, communities and cultures.

- [SC.8.PE.1.1:](#) Use an expression for a specified purpose.

Examples:

Example: Simon has been asked to develop a rock/paper/scissors game. He assigns rock as the number 1, paper as the number 2 and scissors as the number 3. Develop expressions that will allow the game to output a winner based off the combination of user and computer selections.

- [SC.8.PE.1.2:](#) Create a programming process for decomposing a problem.

Clarifications:

Clarification 1: Within this benchmark, the term function and procedure can be used interchangeably.

Clarification 2: Instruction includes consideration of the intake and output of information.

- [SC.8.PE.1.3:](#) Create a function with parameters.

Clarifications:

Clarification 1: Instruction includes understanding that a parameter is a kind of variable that is defined in the function.

- [SC.8.PE.1.4:](#) Explain the use of iterative and non-iterative structures and their uses as a code segment.

Clarifications:

Clarification 1: Instruction includes discussing that iteration is a process repeated until a specific end result.

- [SC.8.PE.1.5:](#) Create an algorithm to solve one or more parts of a decomposed problem.

Clarifications:

Clarification 1: Instruction focuses on creating algorithms that are efficient, reliable and valid.

Clarification 2: Context for problems include video games, robot obstacle course and making dinner.

- [SC.8.PE.1.6:](#) Create an algorithm that can collect data.
- [SC.8.PE.1.7:](#) Design an application for a specified purpose.

Examples:

Example: A marine biologist conducting research at the FSU Coastal and Marine Laboratory is trying to determine why periwinkle

snails climb to the top of marsh seagrass. Design a program to determine the time of day that the majority of snails climb the seagrass. Analyze the data to determine what events cause the snails to climb at that certain time.

- [SC.8.PE.1.8](#): Recognize different numerical data types.

Clarifications:

Clarification 1: Instruction includes decimal type (floating point) and integers.

- [SC.8.PE.1.9](#): Design a program that will assist a user with equations using standard mathematical operators.

Clarifications:

Clarification 1: Instruction includes the use of addition, subtraction, multiplication and division.

Clarification 2: Within this benchmark, instruction includes the use of inequalities.

Clarification 3: Instruction includes flowcharting the initial sequence of steps.

- [SC.8.PE.1.10](#): Create a code segment using iteration.
- [SC.8.PE.1.11](#): Identify the limitations that need to be recognized when creating an algorithm.

Clarifications:

Clarification 1: Instruction includes the necessity to abide by mathematical rules.

- [SC.8.PE.1.12](#): Select an efficient algorithm for a given task based on certain criteria.

Clarifications:

Clarification 1: Within this benchmark, the expectation is that students will not create algorithms, a list will be provided by the teacher.

Clarification 2: Criteria may include time, resources and accessibility.

- [SC.8.PE.2.1](#): Select and use applicable data-collection technology.

Clarifications:

Clarification 1: Instruction includes gathering, viewing, organizing and analyzing data.

Clarification 2: Data-collection tools include probes, handheld devices, geographic mapping systems and output from multiple runs of a computer program.

- [SC.8.PE.2.2:](#) Utilize data-collection technology to report results for content-related problems.

Clarifications:

Clarification 1: Instruction includes allowing students to operate individually and collaboratively.

- [SC.8.PE.2.3:](#) Utilize data from simulations to test hypotheses.

Clarifications:

Clarification 1: Instruction includes the use of digital modeling.

- [SC.8.PE.2.4:](#) Perform a variety of operations such as sorting, filtering and searching in a database.
- [SC.8.PE.2.5:](#) Utilize organized data within a database to solve a problem.

Clarifications:

Clarification 1: Instruction includes the selection and display of the data using an appropriate graph.

- [SC.8.PE.3.1:](#) Represent natural phenomena using a model.
- [SC.8.PE.3.2:](#) Explore the purpose of a class.

Clarifications:

Clarification 1: Instruction includes understanding that classes can be used to create new objects when programming.

- [SC.8.PE.3.3:](#) Evaluate the benefits and limitations of the use of models.

Clarifications:

Clarification 1: Instruction includes consideration of models.

Clarification 2: Within this benchmark, instruction includes consideration of safety, cost, time, location and precision.

Examples:

Example: Mrs. Hooper's class is studying the effects of erosion on

the Florida Gulf coastline. Her class is located in Duval County and is not adjacent to the Gulf coast. Evaluate the benefits and limitations of developing a model to simulate the effects of erosion on the Gulf Coast of Florida.

- [SC.8.PE.4.1:](#) Explore the purpose of the software development life cycle.
- [SC.8.PE.4.2:](#) Explain the phases of a simple software development life cycle.

Clarifications:

Clarification 1: Instruction includes the following phases: describe the project, list necessary steps, take resources into consideration, create a visual representation, actualize the code then perform maintenance for needed changes.

- [SC.8.PE.4.3:](#) Discuss the role of maintenance in the software development cycle.

Clarifications:

Clarification 1: Instruction includes the outcome of changing, modifying and improving the project to meet the user's needs.

- [SC.8.TI.2.1:](#) Describe legal and ethical behaviors when using technology.
- [SC.K12.CTR.1.1:](#) **Actively participate in effortful learning both individually and collaboratively.**
Students who actively participate in effortful learning both individually and with others:
 - Build perseverance by modifying methods as needed while solving a challenging task.
 - Stay engaged and maintain a positive mindset when working to solve tasks.
 - Help and support each other when attempting a new method or approach.

Clarifications:

Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of learners.
- Foster perseverance in students by choosing challenging tasks.
- Recognize students' effort when solving challenging problems.
- Emphasize project-based learning.

- Establish a culture in which students ask questions of the teacher and their peers, and errors as a learning opportunity.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

- **SC.K12.CTR.2.1: Demonstrate understanding by decomposing a problem.**

Students who demonstrate understanding by decomposing a problem:

- Analyze the problems in a way that makes sense given the task.
- Ask questions that will help with solving the task.
- Break down complex problems into individual problems.
- Decompose a complex problem into manageable parts.

Clarifications:

Teachers who encourage students to demonstrate understanding by decomposing a problem:

- Develop students' ability to analyze and problem-solve.
- Help students break complex tasks into subtasks.
- Show students that the solution to individual parts allows them to solve complex problems more effectively.

- **SC.K12.CTR.3.1:**

Complete tasks with digital fluency.

Students who complete tasks with digital fluency: Select and use appropriate digital tools by their functions.

- Demonstrate proper typing techniques and keyboarding skills.
- Understand responsible technology use.
- Use feedback to improve efficiency using digital tools.
- Relate previously learned concepts to new concepts.
- Solve problems by developing, testing and refining technological processes.

Clarifications:

Teachers who encourage students to complete tasks with digital fluency:

- Provide students with opportunities to increase critical thinking skills.
- Provide students with opportunities to use various technology hardware and software, so that technology is an integral part of the learning experience. Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.
- [SC.K12.CTR.4.1:](#)

Express solutions as computational steps.

Students who express solutions as computational steps:

- Solve problems step by step rather than all at once.
- Represent solutions to problems in multiple ways, based on context or purpose.
- Use patterns and structures to understand and connect computational concepts.
- Check computations when solving problems.

Clarifications:

Teachers who encourage students to express solutions as computational steps:

- Provide opportunities for students to develop sequentially based understandings of problems.
- Guide students to align tasks to a step-by-step solution.
- Select sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Prompt students to continually ask, "Does this solution make sense? How do you know?"
- Reinforce that students check their work as they progress within and after a task.
- Strengthen students' ability to verify solutions through justification.
- [SC.K12.CTR.5.1:](#)

Create an algorithm to achieve a given goal.

Students who create algorithms to achieve a given goal:

- Create or use a well-defined series of steps to achieve a desired outcome.

- Compare the efficiency of an algorithm to those expressed by others.
- Design a sequence of steps to follow.
- Verify possible solutions by explaining the program or methods used.

Clarifications:

Teachers who encourage students to create an algorithm to achieve a given goal:

- Support students to develop generalizations based on the similarities found among problems.
- Have students estimate or predict solutions before solving.
- Help students recognize the patterns in the world around them and connect these patterns to other concepts.

• [SC.K12.CTR.6.1:](#)

Differentiate between usable data and miscellaneous information.

Students who differentiate between usable data and miscellaneous information:

- Express connections between concepts and representations.
- Construct possible arguments based on evidence.
- Perform decision-making between two actions.
- Practice evaluating information and sources.
- Perform investigations to gather data or determine if a program or method is appropriate.
- Discern relevant, meaningful data from irrelevant or extraneous information.
- Understand the characteristics and criteria determining whether data is relevant to a specific problem or task.

Clarifications:

Teachers who encourage students to differentiate between useable data and miscellaneous information:

- Support students as they validate conclusions by comparing them to the given situation.
- Create opportunities for students to discuss their thinking with peers.

- SC.K12.CTR.7.1:

Solve real-life problems in science and engineering using computational thinking.

Students who solve real-life problems in science and engineering using computational thinking:

- Adapt procedures to find solutions and apply them to a new context.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.
- Connect concepts to everyday experiences.
- Use programs, models and methods to understand, represent and solve problems.
- Indicate how various concepts can be applied to other disciplines.
- Redesign programs, models and methods to improve accuracy or efficiency. Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to solve real-life problems in science and engineering using computational thinking:

- Create learning opportunities that require logical reasoning and problem-solving skills.
- Provide opportunities for students to create plans and procedures to solve problems.
- Provide opportunities for students to create programs or models, both concrete and abstract, and perform investigations.
- Challenge students to question the accuracy of their programs, models and methods.

- MA.K12.MTR.1.1: **Actively participate in effortful learning both individually and collectively.**

Mathematicians who participate in effortful learning both individually and with others:

- Analyze the problem in a way that makes sense given the task.
- Ask questions that will help with solving the task.

- Build perseverance by modifying methods as needed while solving a challenging task.
- Stay engaged and maintain a positive mindset when working to solve tasks.
- Help and support each other when attempting a new method or approach.

Clarifications:

Teachers who encourage students to participate actively in effortful learning both individually and with others:

- Cultivate a community of growth mindset learners.
- Foster perseverance in students by choosing tasks that are challenging.
- Develop students' ability to analyze and problem solve.
- Recognize students' effort when solving challenging problems.

• **MA.K12.MTR.2.1: Demonstrate understanding by representing problems in multiple ways.**

Mathematicians who demonstrate understanding by representing problems in multiple ways:

- Build understanding through modeling and using manipulatives.
- Represent solutions to problems in multiple ways using objects, drawings, tables, graphs and equations.
- Progress from modeling problems with objects and drawings to using algorithms and equations.
- Express connections between concepts and representations.
- Choose a representation based on the given context or purpose.

Clarifications:

Teachers who encourage students to demonstrate understanding by representing problems in multiple ways:

- Help students make connections between concepts and representations.
- Provide opportunities for students to use manipulatives when investigating concepts.

- Guide students from concrete to pictorial to abstract representations as understanding progresses.
- Show students that various representations can have different purposes and can be useful in different situations.

- **MA.K12.MTR.3.1: Complete tasks with mathematical fluency.**

Mathematicians who complete tasks with mathematical fluency:

- Select efficient and appropriate methods for solving problems within the given context.
- Maintain flexibility and accuracy while performing procedures and mental calculations.
- Complete tasks accurately and with confidence.
- Adapt procedures to apply them to a new context.
- Use feedback to improve efficiency when performing calculations.

Clarifications:

Teachers who encourage students to complete tasks with mathematical fluency:

- Provide students with the flexibility to solve problems by selecting a procedure that allows them to solve efficiently and accurately.
- Offer multiple opportunities for students to practice efficient and generalizable methods.
- Provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.

- **MA.K12.MTR.4.1: Engage in discussions that reflect on the mathematical thinking of self and others.**

Mathematicians who engage in discussions that reflect on the mathematical thinking of self and others:

- Communicate mathematical ideas, vocabulary and methods effectively.
- Analyze the mathematical thinking of others.
- Compare the efficiency of a method to those expressed by others.
- Recognize errors and suggest how to correctly solve the task.
- Justify results by explaining methods and processes.

- Construct possible arguments based on evidence.

Clarifications:

Teachers who encourage students to engage in discussions that reflect on the mathematical thinking of self and others:

- Establish a culture in which students ask questions of the teacher and their peers, and error is an opportunity for learning.
- Create opportunities for students to discuss their thinking with peers.
- Select, sequence and present student work to advance and deepen understanding of correct and increasingly efficient methods.
- Develop students' ability to justify methods and compare their responses to the responses of their peers.

- **MA.K12.MTR.5.1: Use patterns and structure to help understand and connect mathematical concepts.**

Mathematicians who use patterns and structure to help understand and connect mathematical concepts:

- Focus on relevant details within a problem.
- Create plans and procedures to logically order events, steps or ideas to solve problems.
- Decompose a complex problem into manageable parts.
- Relate previously learned concepts to new concepts.
- Look for similarities among problems.
- Connect solutions of problems to more complicated large-scale situations.

Clarifications:

Teachers who encourage students to use patterns and structure to help understand and connect mathematical concepts:

- Help students recognize the patterns in the world around them and connect these patterns to mathematical concepts.
- Support students to develop generalizations based on the similarities found among problems.
- Provide opportunities for students to create plans and procedures to solve problems.

- Develop students' ability to construct relationships between their current understanding and more sophisticated ways of thinking.

- **MA.K12.MTR.6.1: Assess the reasonableness of solutions.**

Mathematicians who assess the reasonableness of solutions:

- Estimate to discover possible solutions.
- Use benchmark quantities to determine if a solution makes sense.
- Check calculations when solving problems.
- Verify possible solutions by explaining the methods used.
- Evaluate results based on the given context.

Clarifications:

Teachers who encourage students to assess the reasonableness of solutions:

- Have students estimate or predict solutions prior to solving.
- Prompt students to continually ask, "Does this solution make sense? How do you know?"
- Reinforce that students check their work as they progress within and after a task.
- Strengthen students' ability to verify solutions through justifications.

- **MA.K12.MTR.7.1: Apply mathematics to real-world contexts.**

Mathematicians who apply mathematics to real-world contexts:

- Connect mathematical concepts to everyday experiences.
- Use models and methods to understand, represent and solve problems.
- Perform investigations to gather data or determine if a method is appropriate.
- Redesign models and methods to improve accuracy or efficiency.

Clarifications:

Teachers who encourage students to apply mathematics to real-world contexts:

- Provide opportunities for students to create models, both concrete and abstract, and perform investigations.

- Challenge students to question the accuracy of their models and methods.
- Support students as they validate conclusions by comparing them to the given situation.
- Indicate how various concepts can be applied to other disciplines.
- [ELA.K12.EE.1.1](#): Cite evidence to explain and justify reasoning.

Clarifications:

K-1 Students include textual evidence in their oral communication with guidance and support from adults. The evidence can consist of details from the text without naming the text. During 1st grade, students learn how to incorporate the evidence in their writing.

2-3 Students include relevant textual evidence in their written and oral communication. Students should name the text when they refer to it. In 3rd grade, students should use a combination of direct and indirect citations.

4-5 Students continue with previous skills and reference comments made by speakers and peers. Students cite texts that they've directly quoted, paraphrased, or used for information. When writing, students will use the form of citation dictated by the instructor or the style guide referenced by the instructor.

6-8 Students continue with previous skills and use a style guide to create a proper citation.

9-12 Students continue with previous skills and should be aware of existing style guides and the ways in which they differ.

- [ELA.K12.EE.2.1](#): Read and comprehend grade-level complex texts proficiently.

Clarifications:

See [Text Complexity](#) for grade-level complexity bands and a text complexity rubric.

- [ELA.K12.EE.3.1](#): Make inferences to support comprehension.

Clarifications:

Students will make inferences before the words infer or inference are introduced. Kindergarten students will answer questions like "Why is the girl smiling?" or make predictions about what will happen based on the title page. Students will use the terms and apply them in 2nd grade and beyond.

- [ELA.K12.EE.4.1](#): Use appropriate collaborative techniques and active listening skills when engaging in discussions in a variety of situations.

Clarifications:

In kindergarten, students learn to listen to one another respectfully.

In grades 1-2, students build upon these skills by justifying what they are thinking. For example: “I think _____ because _____.” The collaborative conversations are becoming academic conversations.

In grades 3-12, students engage in academic conversations discussing claims and justifying their reasoning, refining and applying skills. Students build on ideas, propel the conversation, and support claims and counterclaims with evidence.

- [ELA.K12.EE.5.1](#): Use the accepted rules governing a specific format to create quality work.

Clarifications:

Students will incorporate skills learned into work products to produce quality work. For students to incorporate these skills appropriately, they must receive instruction. A 3rd grade student creating a poster board display must have instruction in how to effectively present information to do quality work.

- [ELA.K12.EE.6.1](#): Use appropriate voice and tone when speaking or writing.

Clarifications:

In kindergarten and 1st grade, students learn the difference between formal and informal language. For example, the way we talk to our friends differs from the way we speak to adults. In 2nd grade and beyond, students practice appropriate social and academic language to discuss texts.

- [ELD.K12.ELL.MA.1](#): English language learners communicate information, ideas and concepts necessary for academic success in the content area of Mathematics.

Course Information

General Notes

This course should be taught using Florida's State Academic Standards for Computer Science: Florida's B.E.S.T. ELA Expectations (EE), Mathematical Thinking and Reasoning Standards (MTRs) and Computational Thinking and Reasoning Standards (CTRs) for students. Florida educators should intentionally embed these standards within the content and their instruction as applicable.

English Language Development ELD Standards Special Notes Section

Teachers are required to provide listening, speaking, reading and writing instruction that allows English language learners (ELL) to communicate information, ideas and concepts for academic success in the content area of Mathematics. For the given level of English language proficiency and with visual, graphic, or interactive support, students will interact with grade level words, expressions, sentences and discourse to process or produce language necessary for academic success. The ELD standard should specify a relevant content area concept or topic of study chosen by curriculum developers and teachers which maximizes an ELL's need for communication and social skills. To access an ELL supporting document which delineates performance definitions and descriptors, please click on the following link: <https://cpalmsmediaproduct.blob.core.windows.net/uploads/docs/standards/eld/si.pdf>.

Accommodations

Federal and state legislation requires the provision of accommodations for students with disabilities as identified on the secondary student's Individual Educational Plan (IEP) or 504 plan or postsecondary student's accommodations' plan to meet individual needs and ensure equal access. Accommodations change the way the student is instructed. Students with disabilities may need accommodations in such areas as instructional methods and materials, assignments and assessments, time demands and schedules, learning environment, assistive technology and special communication systems. Documentation of the accommodations requested and provided should be maintained in a confidential file.

In addition to accommodations, some secondary students with disabilities (students with an IEP served in Exceptional Student Education (ESE)) will need modifications to meet their needs. Modifications change the outcomes and or what the student is expected to learn, e.g., modifying the curriculum of a secondary career and technical education course.

General Information

Course Number: 0200023

Course Path: **Section:** Grades PreK to 12 Education Courses >

Grade Group: Grades 6 to 8 Education Courses > **Subject:**

Computer Science > **SubSubject:** General >

Abbreviated Title: M/J GR8 DIG DISC

Course Length: Semester (S)

Course Type: Elective Course

Course Level: 2

Course Status: Course Approved

Educator Certifications

One of these educator certification options is required to teach this course.

- [Computer Science \(Elementary and Secondary Grades K-12\)](#)
- [Classical Education - Restricted \(Elementary and Secondary Grades K-12\)](#)

Section 1012.55(5), F.S., authorizes the issuance of a classical education teaching certificate, upon the request of a classical school, to any applicant who fulfills the requirements of s. 1012.56(2)(a)-(f) and (11), F.S., and Rule 6A-4.004, F.A.C. Classical schools must meet the requirements outlined in s. 1012.55(5), F.S., and be listed in the [FLDOE Master School ID](#) database, to request a restricted classical education teaching certificate on behalf of an applicant.

Qualifications

As well as any certification requirements listed on the course description, the following qualifications may also be acceptable for the course:

Any field when certification reflects a bachelor or higher degree.